

A Short Guide to Proposition C.4

The orbital Gaussian flat-prior upper bound

Abstract

Proposition C.4 proves that no channel can amplify the displaced thermal Gaussian model with asymptotic flat-prior fidelity larger than $B_{\text{orb}}(\gamma, p)$. Its proof averages a channel over a large phase-space cube, extracts an exactly covariant trace-nonincreasing limit using only compact observables, and evaluates a specially chosen weighted moment. The compact witness avoids any need to pass fidelity itself through the weak limit. This note explains the argument and how Lemma C.1, Lemma C.2, and Proposition C.3 fit together.

1 The Gaussian problem

The orbital LAN model consists of displaced product thermal states on a finite-mode Fock space $\mathcal{F}_{\text{orb}}$:

$$\Phi_z = D(z)\tau D(z)^*, \quad \Psi_z = D(\sqrt{\gamma}z)\tau D(\sqrt{\gamma}z)^*.$$

The notation suppresses that the target has the same centered thermal state τ as the ideal product model, whereas a physical amplifier necessarily produces the noisier centered state $\tau^{(\gamma)}$. The product quantum-limited amplifier sends

$$\Phi_z \mapsto D(\sqrt{\gamma}z)\tau^{(\gamma)} D(\sqrt{\gamma}z)^*.$$

Its fidelity with Ψ_z is independent of z and equals

$$B(\tau^{(\gamma)}, \tau) = B_{\text{orb}}(\gamma, p).$$

For the cube

$$Z_L = \{z : |\Re z_\ell|, |\Im z_\ell| \leq L\},$$

Proposition C.4 supplies the converse inequality

$$\limsup_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|Z_L|} \int_{Z_L} B(\Lambda(\Phi_z), \Psi_z) dz \leq B_{\text{orb}}(\gamma, p). \quad (1)$$

2 The witness and its three properties

Write the two centered thermal states in their joint number basis:

$$\tau = \sum_{\mathbf{k}} t_{\mathbf{k}} |\mathbf{k}\rangle\langle\mathbf{k}|, \quad \tau^{(\gamma)} = \sum_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)} |\mathbf{k}\rangle\langle\mathbf{k}|.$$

Define

$$W_\gamma = \sum_{\mathbf{k}} \sqrt{\frac{t_{\mathbf{k}}}{t_{\mathbf{k}}^{(\gamma)}}} |\mathbf{k}\rangle\langle\mathbf{k}|. \quad (2)$$

This witness is chosen so that three otherwise separate parts of the proof match exactly.

1. It is bounded, positive, injective, and compact. Its eigenvalues are products of decreasing geometric sequences and tend to zero at high total photon number.
2. It is an admissible product of decreasing photon-number weights for the least-noise moment in Proposition C.3.
3. It satisfies

$$\text{Tr}(\tau^{(\gamma)} W_\gamma) = \text{Tr}(\tau W_\gamma^{-1}) = \sum_{\mathbf{k}} \sqrt{t_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)}} = B_{\text{orb}}(\gamma, p). \quad (3)$$

Lemma C.1, the weighted fidelity inequality, therefore gives for every positive trace-class R

$$B(R, \tau)^2 \leq \text{Tr}(R W_\gamma) B_{\text{orb}}(\gamma, p). \quad (4)$$

The nonlinear fidelity has been reduced to one linear compact-observable moment.

3 Statement of Proposition C.4

Proposition C.4 (rephrased). *For the orbital Gaussian input and target families,*

$$\limsup_{L \rightarrow \infty} \sup_{\Lambda} \int_{Z_L} B(\Lambda(\Phi_z), \Psi_z) \frac{dz}{|Z_L|} \leq B_{\text{orb}}(\gamma, p),$$

where the supremum is over all channels on the orbital Fock space.

4 Proof walkthrough

Step 1: move every displacement back to the origin

For a channel Λ , define its finite Følner average

$$\bar{\Lambda}_L = \frac{1}{|Z_L|} \int_{Z_L} \text{Ad}_{D(-\sqrt{\gamma}z)} \circ \Lambda \circ \text{Ad}_{D(z)} \, dz. \quad (5)$$

It is again a channel. Joint concavity and unitary invariance of root fidelity imply

$$\frac{1}{|Z_L|} \int_{Z_L} B(\Lambda(\Phi_z), \Psi_z) \, dz \leq B(\bar{\Lambda}_L(\tau), \tau). \quad (6)$$

Thus all parameter values are collected into the performance of one averaged channel at the centered input.

Step 2: the averaged maps become covariant

Choose $L_j \rightarrow \infty$ and channels whose payoffs approach the limsup in (1); write $\Gamma_j = \bar{\Lambda}_{L_j}$. Shifting the integration variable in (5) gives, for every fixed w ,

$$\begin{aligned} & \|\Gamma_j(D(w)X D(w)^*) - D(\sqrt{\gamma}w)\Gamma_j(X)D(\sqrt{\gamma}w)^*\|_1 \\ & \leq \frac{|(Z_{L_j} + w) \triangle Z_{L_j}|}{|Z_{L_j}|} \|X\|_1. \end{aligned} \quad (7)$$

Only the boundary portion on which the shifted and unshifted cubes differ contributes. Euclidean cubes are a Følner family, so the ratio on the right tends to zero.

Lemma C.2 now yields a subsequence and a completely positive trace-nonincreasing map Γ such that

$$\mathrm{Tr}[\Gamma_j(X)K] \longrightarrow \mathrm{Tr}[\Gamma(X)K] \quad (K \text{ compact}). \quad (8)$$

Equation (7) also makes the limit exactly displacement covariant with gain $\sqrt{\gamma}$.

The limit may lose trace because oscillator mass can escape to high energy, but this causes no problem: Proposition C.3 was formulated for trace-nonincreasing maps.

Step 3: pass the compact witness moment to the limit

Apply (4) to $R = \Gamma_j(\tau)$:

$$\mathrm{B}(\Gamma_j(\tau), \tau)^2 \leq \mathrm{Tr}[\Gamma_j(\tau)W_\gamma] B_{\mathrm{orb}}(\gamma, p).$$

Since W_γ is compact, (8) gives

$$\mathrm{Tr}[\Gamma_j(\tau)W_\gamma] \longrightarrow \mathrm{Tr}[\Gamma(\tau)W_\gamma]. \quad (9)$$

This is the key reason for introducing the weighted witness before taking a limit. One never needs an upper-semicontinuity theorem for fidelity in the chosen topology.

Step 4: use the least-noise moment

The limiting map has exactly the hypotheses of Proposition C.3. Therefore

$$\begin{aligned} \mathrm{Tr}[\Gamma(\tau)W_\gamma] &\leq \mathrm{Tr}[\Gamma(\tau)] \mathrm{Tr}[\tau^{(\gamma)}W_\gamma] \\ &\leq B_{\mathrm{orb}}(\gamma, p), \end{aligned}$$

where the last line uses $\mathrm{Tr} \Gamma(\tau) \leq 1$ and (3). Combining this with (4)–(9) gives

$$\limsup_j \mathrm{B}(\Gamma_j(\tau), \tau)^2 \leq B_{\mathrm{orb}}(\gamma, p)^2.$$

Root fidelity is nonnegative, so taking square roots and returning to (6) proves (1).

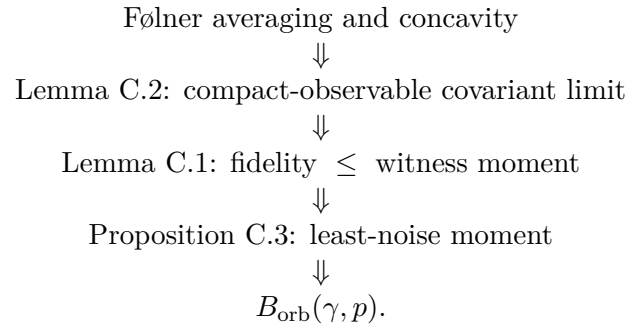
5 Why the argument is sharp

The witness is not an arbitrary estimate. Its eigenvalue at $\mathbf{k}$ is the square-root likelihood ratio between the ideal thermal law and the quantum-limited output law. Consequently, both factors in the weighted Cauchy–Schwarz bound equal their classical Hellinger overlap. The least-noise proposition then says that no admissible covariant limit can have a larger expectation of this witness.

The product quantum-limited amplifier attains $B_{\mathrm{orb}}(\gamma, p)$ at every z , so Proposition C.4 supplies the matching upper bound used in the Gaussian flat-prior theorem.

6 Logical dependencies

The proof has the compact chain



The proof is entirely orbital: there is no classical spectral coordinate. Proposition C.6 repeats this architecture with an additional classical witness and convolution argument.